E-Commerce Sales Analysis Report

1. Project Overview

This project analyzes an e-commerce fashion sales dataset using PostgreSQL to identify business patterns across sales channels, customer demographics, product categories, geography, order status and order types.

The analysis uses SQL aggregation, CASE expressions, window functions, date functions and conditional aggregation to convert transaction-level data into business-facing metrics.

2. Business Objectives

- Measure revenue and order contribution by sales channel.
- Compare customer purchasing patterns by gender and age group.
- Identify high-performing states and broad geographic regions.
- Find the strongest product categories and most frequently ordered sizes.
- Measure cancellation and return incidence by channel.
- Understand monthly sales patterns, B2B versus B2C contribution, and delivery-status distribution.

3. Technical skills demonstrated

- SQL aggregations
- Window functions
- CASE WHEN statements
- Conditional aggregation
- Date and time functions
- Customer segmentation
- Business KPI analysis
- Data interpretation

4. Business Questions:

1. Sales Channel Performance and Revenue Contribution

```sql
-- "Amazon and Flipkart together account for the majority
--  of total revenue compared to other channels."

select
     channel,
     count(*) as total_orders,
     sum(amount) as total_revenue,
     ROUND(SUM(amount) * 100.0 / SUM(SUM(amount)) OVER (),2) as revenue_share_pct
from  dataset
group by channel
ORDER BY total_revenue DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No: 1 of 1

	channel text	total_orders bigint	total_revenue bigint	revenue_share_pct numeric
1	Amazon	11006	7512955	35.48
2	Myntra	7236	4927415	23.27
3	Flipkart	6703	4573301	21.60
4	Ajio	1931	1331427	6.29
5	Nalli	1484	1015329	4.79
6	Meesho	1397	926973	4.38
7	Others	1261	867241	4.10
8	Menyntra	18	14125	0.07
9	AMenaz...	10	6978	0.03
10	Meneesho	1	633	0.00

Key insights:

- Amazon is the largest sales channel, contributing 35.48% of total revenue.
- Myntra and Flipkart are the second and third largest contributors.
- Amazon and Flipkart together contribute approximately 57.08% of total revenue.
- Smaller channels contribute relatively less revenue.

Business recommendation:

The business should prioritize its relationship with high-performing channels while exploring opportunities to improve sales through smaller platforms.

2. Gender-Based Sales Analysis

Query | Query History

```
39  -- Women generate significantly more revenue than Men.
40
41  select gender,
42       count(*) as total_orders,
43       sum(amount) as total_revenue ,
44       ROUND(avg(amount) , 2) as avg_order_value
45  from dataset
46  group by gender
47  order by total_revenue DESC ;
48
```

Data Output | Messages | Notifications

Showing rows: 1 to 2 | Page No: 1 of 1

	gender text	total_orders bigint	total_revenue bigint	avg_order_value numeric
1	Women	21553	13562773	629.28
2	Men	9494	7613604	801.94

Key insights:

- Women contribute approximately 64% of total revenue.
- Women generate significantly more orders than men.
- Men have a higher average order value (AOV) of 801.94 compared with 629.28 for women.
- Women's higher overall revenue is primarily associated with their greater order volume.

Business recommendation:

Develop separate marketing strategies for both segments. Focus on increasing men's purchase frequency while maintaining strong engagement with women customers.

3. Age Group Analysis

```sql
---The 26-35 age group (young adults) places the most orders.

select
    case
       when age between 18 and 25 then '18-25'
       when age between  26 and 35 then '26-35'
       when age between 36 and 45 then '36-45'
       when age between 46 and 55 then '46-55'
       when age between 56 and 65 then '56-65'
       else '66+'
    end as age_group,
    count(*) as total_orders,
    sum(amount ) as total_revenue,
    round(avg(amount), 2) as avg_order_value
from dataset
group by age_group
order by total_orders DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1 of 1

	age_group text	total_orders bigint	total_revenue bigint	avg_order_value numeric
1	26-35	7850	5352829	681.89
2	36-45	7793	5314130	681.91
3	18-25	6206	4221619	680.25
4	46-55	4354	2983560	685.25
5	66+	2743	1886910	687.90

Key insights:

- The 26–35 age group records the highest order volume.
- The 36–45 age group is very close, with only 57 fewer orders.
- Both groups generate approximately 5.3 million in revenue.
- The 66+ segment has the highest AOV among the displayed groups, despite having fewer orders.

Business recommendation:

Focus customer acquisition campaigns on customers aged 26–45 because they represent the strongest order-volume segments.

4. State-wise Sales Performance

Query | Query History

```
70  -- Find  the top-performing state by revenue.
71  select
72      ship_state,
73      count(*) as total_orders,
74      sum(amount) as total_revenue
75  from dataset
76  group by ship_state
77  order by total_revenue DESC;
```

Data Output | Messages | Notifications

SQL

Showing rows: 1 to 53 | Page No: 1 of 1

	ship_state text	total_orders bigint	total_revenue bigint
1	MAHARASHTRA	4509	2982659
2	KARNATAKA	3988	2646358
3	UTTAR PRADESH	2882	2104659
4	TELANGANA	2527	1712439
5	TAMIL NADU	2678	1678244
6	DELHI	1804	1248111
7	KERALA	1549	1008940
8	WEST BENGAL	1425	922444
9	ANDHRA PRADESH	1369	918499
10	HARYANA	1118	813320
11	GUJARAT	1058	715563
12	MADHYA PRADESH	767	564026

Key insights:

- Maharashtra is the highest-performing state in terms of orders and revenue.
- Karnataka holds the second position.
- Uttar Pradesh is the third-largest revenue contributor in the displayed results.
- Maharashtra and Karnataka together generate approximately 5.63 million in revenue.

Business recommendation:

Prioritize inventory availability and regional marketing in Maharashtra and Karnataka while investigating opportunities to expand in other states.

5. Product Category Performance

Query | Query History

```sql
--- Find the best-selling category by volume and revenue.

select
    category,
    count(*) as total_orders,
    sum(qty) as total_units_sold,
    sum(amount) as total_revenue,
    round(avg(amount) , 2) as avg_selling_price
from dataset
group by category
order by total_revenue DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 8 | Page No: 1 of 1

	category text	total_orders bigint	total_units_sold bigint	total_revenue bigint	avg_selling_price numeric
1	Set	12391	12446	10507546	848.00
2	kurta	10446	10541	4959377	474.76
3	Western Dress	4066	4084	3148836	774.43
4	Top	2193	2201	1186199	540.90
5	Saree	1380	1389	1010471	732.23
6	Ethnic Dress	264	264	195256	739.61
7	Blouse	229	234	140888	615.23
8	Bottom	78	78	27804	356.46

Key insights:

- Set is the best-performing category in both order volume and revenue.
- Set generates more than twice the revenue of Kurta.
- Kurta is the second-largest category by revenue.
- Bottom has the lowest order volume among the displayed categories.

Business recommendation:

Prioritize stock availability for Set and Kurta categories. Evaluate product margins before making inventory investment decisions.

6. Cancellation and Return Rate by Channel

```sql
--- Find the cancellation and return rate by channel.
select
    channel,
    count(*) as total_orders,
    sum(case when status ='Cancelled' then 1 else 0 end) as cancelled,
    sum(case when status ='Returned' then 1 else 0 end) as returned,
    round((sum(case when status in ('Cancelled' ,'Returned') then 1 else 0 end)
    *100.0) / count(*), 2) as cancel_return_rate_pct
from dataset
group by channel
order by cancel_return_rate_pct DESC ;
```

Output Messages Notifications

SQL

ıg rows: 1 to 10 Page No: 1 of 1

channel text	total_orders bigint	cancelled bigint	returned bigint	cancel_return_rate_pct numeric
AMenaz...	10	0	2	20.00
Amazon	11006	308	381	6.26
Flipkart	6703	188	229	6.22
Meesho	1397	38	47	6.08
Ajio	1931	54	60	5.90
Myntra	7236	188	237	5.87
Nalli	1484	34	53	5.86
Others	1261	34	36	5.55
Meneesho	1	0	0	0.00

Key insights:

- Amazon has the highest combined cancellation and return rate among the major channels shown.
- Flipkart follows closely at 6.22%.
- Myntra has a comparatively lower rate of 5.87%.
- Channel performance should be evaluated not only by revenue but also by order quality.

Business recommendation:

Investigate product quality, delivery delays, sizing issues and customer complaints to identify the reasons behind returns and cancellations.

7. Monthly Sales Performance

```sql
-- Find top sales month by revenue and volume .
select
    extract (month from date) as month_num,
    to_char(date, 'Mon YYYY') as month,
    count(*) as total_orders,
    sum(amount) as total_revenue
from dataset
group by month_num , month
order by month_num;
```

Data Output | Messages | Notifications

Showing rows: 1 to 12 | Page No: 1 of 1

	month_num numeric	month text	total_orders bigint	total_revenue bigint
1	1	Jan 2022	2702	1820601
2	2	Feb 2022	2750	1875932
3	3	Mar 20...	2819	1928066
4	4	Apr 2022	2685	1829263
5	5	May 20...	2617	1797822
6	6	Jun 2022	2597	1750966
7	7	Jul 2022	2579	1772300
8	8	Aug 20...	2617	1808505
9	9	Sep 2022	2490	1688871
10	10	Oct 2022	2424	1666662
11	11	Nov 20...	2383	1615356

Key insights:

- March has the highest visible order volume and revenue.
- Revenue decreases after March, with some recovery in July and August.
- November records the lowest visible order volume.
- The business may experience seasonal fluctuations in customer demand.

Business recommendation:

Analyze seasonal purchasing patterns and plan promotional campaigns during months with declining sales.

8. B2B vs. B2C Sales Analysis

```
116
117  ---Find higher order_type by volume and revenue .
118
119  select
120    case when b2b then 'B2B' else 'B2C' end as order_type,
121    count(*) as total_orders,
122    sum(amount) as total_revenue,
123    round(avg(amount) , 2) as avg_order_value,
124    max(amount) as max_order_value
125  from dataset
126  group by b2b;
```

Data Output | Messages | Notifications

Showing rows: 1 to 2 | Page No: 1 of 1

	order_type text	total_orders bigint	total_revenue bigint	avg_order_value numeric	max_order_value integer
1	B2C	30861	21050056	682.09	3036
2	B2B	186	126321	679.15	1523

Key insights:

- B2C contributes the majority of total revenue.
- B2B contributes a very small share of overall transactions.
- Average order values are relatively similar between B2B and B2C customers.

Business recommendation:

Focus on improving B2C customer retention while exploring opportunities to grow the B2B customer base.

9. Most Ordered Product Sizes

```
128
129   -- Find  the most ordered sizes.
130   select
131       size,count(*) as total_orders,
132       sum(amount) as total_revenue
133   from dataset
134   group by size
135   order by  total_orders DESC ;
136
```

Data Output | Messages | Notifications

SQL

Showing rows: 1 to 12 | Page No: 1 of 1

	size text	total_orders bigint	total_revenue bigint
1	M	5479	3768582
2	L	5144	3445476
3	XL	4437	2950033
4	S	4176	2923443
5	XXL	4065	2677640
6	3XL	3347	2259099
7	XS	2503	1774580
8	Free	1408	1022905
9	6XL	208	152198
10	5XL	161	119039
11	4XL	113	79734

Key insights:

- Medium (M) is the most frequently ordered size.
- Large (L) is the second most popular size.
- XL also has substantial customer demand.
- Smaller and extended sizes have comparatively lower order volumes.

Business recommendation:

Use size-level demand analysis to optimize inventory distribution and reduce stockouts or excess inventory.

10. Older Customer Preferences by Category

```sql
-- Find Older customers preferences category-wise and volume .
select
    case
       when age < 35 then 'Under 35'
       when age between 35 and 54 then '35-54'
       else '55+'
     end as age_group,
     category,
     count(*)  as total_orders
from dataset
group by age_group , category
order by age_group , total_orders DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 24 | Page No: 1 of 1

	age_group text	category text	total_orders bigint
1	35-54	Set	5092
2	35-54	kurta	4236
3	35-54	Western Dress	1683
4	35-54	Top	890
5	35-54	Saree	582
6	35-54	Ethnic Dress	115
7	35-54	Blouse	91
8	35-54	Bottom	29
9	55+	Set	2041

Key insights:

- Customers aged 35–54 show the highest demand for Sets.
- Kurta is the second most frequently ordered category among this age group.
- Set is also the leading category among customers aged 55+ in the visible results.
- Traditional and ethnic fashion categories appear important for these customer segments.

Business recommendation:

Develop targeted product recommendations and marketing campaigns for older customer segments based on their category preferences.

11. Order Delivery and Status Distribution

```sql
----Delivered orders make up more than 80% of all orders.
select
    status,
    count(*) as total_orders,
    round(count(*) * 100.0 / sum(count(*)) over() , 2) as pct_of_total
from dataset
group by status
order by total_orders DESC ;
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 | Page No: 1 of 1

	status text	total_orders bigint	pct_of_total numeric
1	Delivered	28641	92.25
2	Returned	1045	3.37
3	Cancelled	844	2.72
4	Refunded	517	1.67

Key insights:

- 92.25% of all orders are successfully delivered.
- Returned orders account for 3.37%.
- Cancelled orders account for 2.72%.
- Refunded orders represent 1.67% of the total.
- Overall, the majority of transactions reach the delivered status.

Business recommendation:

Maintain the current delivery performance while investigating the causes of returns, cancellations and refunds to improve customer satisfaction.

12. Highest-Spending Customer Segments (Gender × Age Group)

```sql
---- Find  the highest-spending customer segment (gender × age group).
select
     gender ,
     case
       when age between 18 and 25 then '18-25'
       when age between 26 and 35 then '26-35'
       when age between 36 and 45 then '36-45'
       when age between 46 and 55 then '46-55'
       else '56+'
     end as age_group,
     count(*) as total_orders,
     sum(amount) as total_revenue,
     round(avg(amount) ,2) as avg_order_value
from dataset
group by gender, age_group
order by total_revenue DESC
LIMIT 5 ;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No: 1 of 1

	gender text	age_group text	total_orders bigint	total_revenue bigint	avg_order_value numeric
1	Women	26-35	5520	3475674	629.65
2	Women	36-45	5328	3344271	627.68
3	Women	18-25	4308	2689824	624.38
4	Women	56+	3369	2133376	633.24
5	Men	36-45	2465	1969859	799.13

Key insights:

- Women aged 26–35 are the largest revenue-generating customer segment in the displayed results.
- Women aged 36–45 are the second-largest segment.
- Women contribute the majority of orders across the displayed top segments.
- Men aged 36–45 have a significantly higher AOV than the displayed women's segments.

Business recommendation:

Prioritize customer retention and personalized campaigns for women aged 26–45. Explore premium product recommendations for men aged 36–45.

13. Regional Sales Performance

```sql
--Find top performing  states in order volume.
select
    case
        when ship_state in  ('KARNATAKA','TAMIL NADU','TELANGANA','ANDHRA PRADESH','KE
            then 'South India'
        when ship_state in ('UTTAR PRADESH','RAJASTHAN','PUNJAB','HARYANA','DELHI',
                            'MADHYA PRADESH','UTTARAKHAND','HIMACHAL PRADESH','BIHAR')
            then 'North India'
        when ship_state in  ('MAHARASHTRA','GOA','GUJARAT')
            then'West India'
        when ship_state in ('WEST BENGAL','ODISHA','JHARKHAND','CHHATTISGARH','ASSAM')
            then 'East/NE India'
        else 'Other'
    end  as region,
    count (*) as total_orders,
    sum(amount)as total_revenue
from dataset
group by region
order by  total_orders DESC;
```

Data Output | Messages | Notifications

SQL

Showing rows: 1 to 5 | Page No: 1 of 1

region text	total_orders bigint	total_revenue bigint
South India	12111	7964480
North India	9043	6552921
West India	5866	3878251

Key insights:

- South India is the leading region in order volume and revenue.
- North India ranks second.
- West India ranks third among the visible results.
- South India generates approximately 34% more orders than North India.

Business recommendation:

Focus on maintaining strong market performance in South India while identifying growth opportunities in North and West India.

EXECUTIVE SUMMARY VIEW :

```
208  ----BONUS: EXECUTIVE SUMMARY VIEW
209  -- Overall KPIs in one shot
210  select
211      count(*)   as total_orders,
212      count(distinct cust_id) as unique_customers,
213      sum(amount) as gross_revenue_inr,
214      round (avg(amount), 2)as avg_order_value,
215      sum(case when status = 'Delivered'  then 1 else 0 end) as delivered,
216      sum(case when status = 'Cancelled'  THEN 1 else 0 end) as cancelled,
217      sum(case when  status = 'Returned'   THEN 1 else 0 end) as returned,
218      sum(case when status = 'Refunded'   THEN 1 else 0 end) as refunded,
219      round(
220          sum(case when status in('Cancelled','Returned') then 1 else 0 end)
221          * 100.0 / count(*), 2
222      )as cancel_return_rate_pct
223  from dataset;
224
```

Data Output | Messages | Notifications

Showing rows: 1 1

	total_orders bigint	unique_customers bigint	gross_revenue_inr bigint	avg_order_value numeric	delivered bigint	cancelled bigint	returned bigint	refunded bigint	cancel_return_rate_pct numeric
1	31047	28437	21176377	682.07	28641	844	1045	517	6.08

- **Total Orders — 31047**
- **Unique Customers — 28437**
- **Gross Revenue — 21176377**
- **Average Order Value — 682.07**
- **Delivered — 28641**
- **Cancelled — 844**
- **Returned — 1045**
- **Refunded — 517**
- **Cancel Return Rate Percentage — 6.08%**